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Performance Improvement of Recommendation Systems using User Evaluation Information

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[Abstract]

Graph Signal Processing (GSP)-based recommender systems have attracted attention due to their high computational efficiency compared with Graph Convolutional Network (GCN)-based approaches. Howeve r, existing GSP-based methods do not explicitly utilize user rating information when constructing interact ion graphs. To address this limitation, this paper proposes a Rating-Aware Graph Filtering method that i ncorporates user ratings into graph filtering. The proposed method was evaluated on five benchmark dat asets: ML-100K, ML-Latest-Small, ML-1M, and CiaoDVD. Experimental results show consistent improve ments over HiGSP. These results demonstrate that incorporating user rating information effectively enhan ces recommendation performance in GSP-based recommender systems.

▸ **Key words** : Recommendation system, Graph signal processing, Graph filtering, User rating, HiGSP

[요 약]

그래프 신호 처리(Graph Signal Processing, GSP) 기반 추천시스템은 그래프 합성곱 신경망(Graph Convolutional Network, GCN) 기반 추천시스템에 비해 높은 연산 효율성을 제공하지만, 사용자 평점 정보를 직접적으로 반영하지 못하는 한계를 가진다. 본 논문에서는 사용자 평점을 반영한 Rati ng-Aware Graph Filtering 기법을 제안하며, 평점 기반 인접행렬과 기존 상호작용 기반 인접행렬을 결합하여 그래프 필터링을 수행한다. 제안 기법의 성능 검증을 위해 ML-100K, ML-Latest-Small, M L-1M, CiaoDVD 데이터셋에서 실험을 수행하였다. 실험 결과, 기존 HiGSP 모델 대비 성능이 향상 되었으며, 이를 통해 사용자 평점 정보를 활용한 그래프 필터링이 추천 성능 향상에 효과적임을 확인하였다.

▸ **주제어** : 추천시스템, 그래프 신호 처리, 그래프 필터링, 사용자 평점, HiGSP

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I. Introduction

With the rapid advancement of the Internet and the widespread adoption of mobile devices, the amount of available information and content has increased dramatically. In particular, as markets in various service domains such as e-commerce, streaming services, and social media continue to expand, the overwhelming abundance of products and content has made it increasingly difficult for users to identify items that best match their preferences. In response to this challenge, recommender systems have emerged as a technology that assists users in making appropriate choices. A recommender system is a type of Information Filtering technology that aims to improve user satisfaction by recommending information that users are likely to find interesting [1][2][3][4]. Therefore, recommender systems contribute to alleviating the problem of information overload while providing significant benefits to both users and service providers.

The two most important factors in recommender systems are training speed and recommendation accuracy. Training speed refers to how quickly a recommender system can learn from newly acquired data and provide users with appropriate recommendations in real time. Since user preferences and behavioral patterns change rapidly over time, a recommender system that fails to promptly reflect these changes may provide outdated recommendations, thereby reducing user satisfaction. In addition, recommendation accuracy is an important metric that indicates how relevant and appropriate the recommended items are for individual users. A highly accurate recommender system effectively captures user preferences and delivers personalized content [5][6][7][8][9][10].

To address these issues, this study proposes a novel method that improves the performance of a Graph Signal Processing (GSP)-based recommender system by incorporating user rating information. Conventional Graph Convolution Network (GCN)-based recommender systems effectively model complex graph structures through graph convolution operations. However, they require a large number of trainable parameters, and the optimization process involves substantial computational cost and long training time [12][13][14][15][16]. As a result, updating the recommender system model is time-consuming. To overcome this limitation, Graph Signal Processing-based recommender systems have been introduced as an efficient alternative for modeling relatively simple relationships [17][18][19][20][21][22].

Graph Signal Processing-based recommender systems perform recommendation through a low-pass filter, which consists of simple matrix operations and does not require the learning of trainable weight parameters. Consequently, the model can be updated efficiently with significantly lower computational overhead. Furthermore, their recommendation accuracy is comparable to or even higher than that of Graph Convolution Network-based recommender systems [19][20][21][22].

However, existing Graph Signal Processing-based recommender systems treat items with high user ratings and those with low user ratings equally without distinguishing between them when constructing the model [22]. Therefore, this study proposes a method that incorporates users' high rating information into the model to achieve more personalized recommendations.

II. Related Works

Recommender systems are generally classified into Content-Based Filtering, Collaborative Filtering, and Hybrid Filtering, which combines the strengths of both approaches [1][2][3][4].

Content-Based Filtering recommends items with characteristics similar to those previously preferred by the user by utilizing item attributes. Although it can generate recommendations without relying on other users' data, it is limited by the difficulty of ex

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tracting item features and its inability to fully capture users' diverse interests.

Collaborative Filtering is based on the assumption that users with similar preferences tend to favor similar items [3][5][6][8][9][10][11]. It is categorized into User-Based and Item-Based Collaborative Filtering [7]. This approach does not require item feature information and can effectively discover users' potential interests. However, it suffers from the Cold Start Problem, the Data Sparsity Problem, and increasing computational complexity as user-item interactions grow.

Hybrid Filtering combines Content-Based and Collaborative Filtering to improve recommendation performance. It can alleviate the Cold Start Problem while enhancing recommendation diversity, although it introduces higher model complexity and computational cost.

With the advancement of deep learning-based Collaborative Filtering, Graph Convolution Networks (GCNs) have been widely applied to recommender systems [12]. Representative models include NGCF [13], LightGCN [14], SimGCL [15], and XSimGCL [16].

Graph Signal Processing (GSP) extends conventional signal processing to graph-structured data by exploiting graph topology [17][18]. Its core concepts include Graph Signals, the Graph Laplacian Matrix, Graph Filtering, and the Graph Fourier Transform (GFT). In particular, Graph Filtering transforms graph signals into the frequency domain, where Low-pass Filters suppress high-frequency components while preserving the similarities among neighboring nodes.

GF-CF (Graph Filter-based Collaborative Filtering) is a representative recommender system based on Graph Signal Processing [19]. To capture user-item similarities, GF-CF employs both Linear Low-pass and Ideal Low-pass Filters. Subsequently, several improved models were proposed, including PGSP, which incorporates richer user information [20], Turbo-CF, which improves computational efficiency [21], and HiGSP, which introduces hierarchical user clustering before graph filtering [22].

This study is based on HiGSP, which achieved state-of-the-art performance among Graph Signal Processing-based recommender systems as of 2024. HiGSP combines a cluster-based filtering module with a global-aware filtering module consisting of an Ideal Low-pass Filter and a High-order Low-pass Filter to effectively model user preferences and predict future interactions.

A previous study [19] compared the Graph Convolution Network-based model LightGCN with the Graph Signal Processing-based model GF-CF. Since GF-CF operates solely through matrix operations without trainable parameters, it achieves significantly higher computational efficiency while maintaining comparable or even superior recommendation performance on the Yelp2018 and Amazon-book datasets. Motivated by these advantages, this study proposes a method that further improves recommendation performance while preserving the computational efficiency of Graph Signal Processing-based recommender systems.

III. The Proposed Scheme

Existing Graph Signal Processing-based recommender systems, including HiGSP, construct the User Interaction Matrix for building the adjacency matrix by considering only whether an interaction exists between a user and an item, without taking user rating information into account [19][20][21][22]. This approach is based on the assumption that users interact more frequently with items they prefer. However, in real-world scenarios, users may repeatedly interact with items to which they assign low ratings, while interacting less frequently with items they rate highly. Consequently, an adjacency matrix constructed solely from interaction information may fail to adequately reflect users' actual preferences, potentially leading to degraded recommendation performance.

To address this limitation, this study proposes a

method that additionally incorporates user rating information during the construction of the User Interaction Matrix. Specifically, a user-item connection is created only when the rating assigned by the user is greater than or equal to a predefined threshold (N) (3 in this study), thereby constructing a new rating-based adjacency matrix. This approach aims to strengthen the signals of users' preferred items and improve personalized recommendation performance.

However, using only the rating-based adjacency matrix may result in information loss because interactions associated with low ratings are excluded. To overcome this issue, this study utilizes both the conventional interaction-based adjacency matrix and the proposed rating-based adjacency matrix. Graph filtering is independently performed on each adjacency matrix, and the resulting representations are combined using a weighted aggregation strategy. This approach effectively incorporates user preference information while minimizing information loss [17][18]. Furthermore, the proposed combination framework provides an ensemble effect. In Chapter 5, the recommendation performance under different threshold values of (N) is compared with that of existing models through comprehensive experiments.

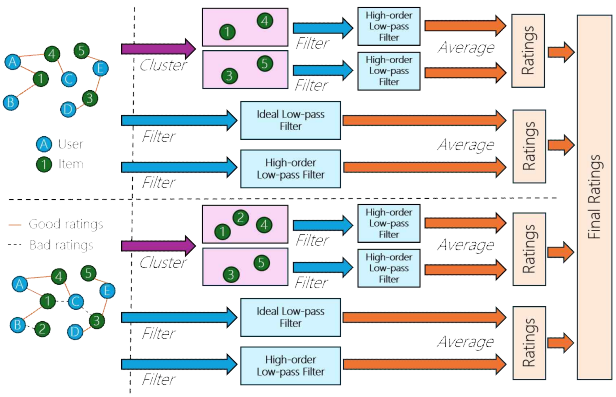


Figure 1. Architecture of the Proposed Model

Figure 1 illustrates the overall architecture of the proposed model. The framework consists of two main components. The lower branch performs graph filtering in the same manner as the original HiGSP model, while the upper branch performs graph filtering using the newly constructed adjacency matrix proposed in this study [22]. By combining these two branches, the proposed model alleviates the information loss problem while achieving more personalized recommendations and benefiting from an ensemble effect.

The lower branch of Figure 1 constructs the adjacency matrix using the same procedure as the original HiGSP model to prevent information loss caused by the proposed method. After constructing the adjacency matrix, clustering is performed to group similar users, followed by the application of a Linear Low-pass Filter to compute the rating scores. Subsequently, the original adjacency matrix is processed using an Ideal Low-pass Filter and a High-order Low-pass Filter, respectively, to generate additional rating scores [22]. Through the application of multiple Low-pass Filters, the model effectively captures user preferences and improves recommendation quality.

The upper branch of Figure 1 constructs a new adjacency matrix using the proposed method and performs graph filtering based on this newly generated graph structure. Similar to the lower branch, clustering is first applied to group users with similar preferences, followed by a Linear Low-pass Filter to compute rating scores. The proposed adjacency matrix is then processed using both an Ideal Low-pass Filter and a High-order Low-pass Filter to generate rating scores. Unlike conventional approaches that rely solely on interaction information, the proposed method incorporates users' high rating information into the graph filtering process, enabling more accurate and personalized recommendations. In the final stage, the four rating matrices generated from the above procedures are combined using weighted aggregation to produce the final rating matrix. Through this framework, the proposed model addresses the limitations of the original HiGSP while simultaneously benefiting from an ensemble effect, thereby improving recommendation performance.

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IV. Experiments

Table 1. Datasets Used in the Experiments

Dataset	ML-100K	ML-Small	ML-1M	Ciao DVD
# of Relations	100K	100K	1M	72K
# of Users	943	610	6,040	17,615
# of Items	1,682	9,724	3,706	16,121

As shown in Table 1, a total of five datasets were used to compare the performance of the existing model and the proposed model. The datasets used in the experiments were ML-100K, ML-1M, ML-Latest-Small, FilmTrust, and CiaoDVD. The MovieLens datasets are representative benchmark datasets for recommender systems provided by GroupLens Research and contain user-movie rating information [23]. CiaoDVD is a dataset containing product reviews and rating information and is widely used for user preference analysis and recommendation performance evaluation [24]. Although ML-100K and ML-1M were used in previous studies, the training and test sets were newly partitioned in this study to enable a more detailed analysis. In addition, ML-Latest-Small, FilmTrust, and CiaoDVD were not used in previous experiments and were therefore employed to further validate the generalization performance of the proposed model. To compare the characteristics of the datasets, the total number of users, total number of items, number of user-item interactions, and density were analyzed and are presented in Table 1. The density of each dataset was calculated using the following equation.

Table 2. Experimental Environment

Resource	Property
OS	Ubuntu 22.04 LTS
CPU	13th Gen Intel(R) Core TM I9-13990K
RAM	DDR 5 32GB + 256G SWAP
GPU	NVIDIA GeForce RTX 4090

The experiments in this study were conducted under the experimental environment described in Table 2 using the package versions listed in Table 3.

Table 3. Package Versions Used in the Experimental Environment

Package	Version
Python	3.7.10
Numpy	1.19.5
Pytorch	1.9.0
Scikit-learn	0.22.0
Scipy	1.5.2
Sparsesvd	0.2.2

Table 4. Hyperparameters Used in the Experiments

Hyperparameter	Setting
alpha1	0.3
alpha2	0.7
order1	8
order2	8
pri_factor	256
n_cluster	5
HiGSP : Ours	5 : 5
random_state	2024

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Table 5. Performance Comparison for Ratings2

Dataset	Metrics	HiGSP	Ours($\geq$2)
ML-100K	F1@5	0.2099	**0.2139 (+1.91%)**
	F1@10	0.2718	**0.2750 (+1.18%)**
	MRR@5	0.5830	**0.5971 (+2.42%)**
	MRR@10	0.5434	**0.5511 (+1.42%)**
	NDCG@5	0.6366	**0.6484 (+1.85%)**
	NDCG@10	0.6182	**0.6226 (+0.71%)**
ML-Latest-Small	F1@5	0.1001	**0.1050 (+4.90%)**
	F1@10	0.1478	**0.1501 (+1.56%)**
	MRR@5	0.3270	**0.3495 (+6.88%)**
	MRR@10	0.3421	**0.3520 (+2.89%)**
	NDCG@5	0.3977	**0.4188 (+5.31%)**
	NDCG@10	0.4439	**0.4523 (+1.89%)**
ML-1M	F1@5	0.1789	**0.1805 (+0.89%)**
	F1@10	0.2457	**0.2462 (+0.20%)**
	MRR@5	0.5984	**0.6000 (+0.27%)**
	MRR@10	**0.5603**	0.5601 (-0.04%)
	NDCG@5	0.6571	**0.6591 (+0.30%)**
	NDCG@10	0.6407	**0.6410 (+0.05%)**
CiaoDVD	F1@5	0.0080	**0.0083 (+3.75%)**
	F1@10	0.0085	**0.0092 (+8.24%)**
	MRR@5	0.0133	**0.0139 (+4.51%)**
	MRR@10	0.0136	**0.0151 (+11.03%)**
	NDCG@5	0.0166	**0.0170 (+2.41%)**
	NDCG@10	0.0204	**0.0223 (+9.31%)**

Table 6. Performance Comparison for Ratings5

Dataset	Metrics	HiGSP	Ours($\geq$5)
ML-100K	F1@5	0.2099	**0.2115 (+0.76%)**
	F1@10	0.2718	**0.2728 (+0.37%)**
	MRR@5	0.5830	**0.5902 (+1.23%)**
	MRR@10	0.5434	**0.5490 (+1.03%)**
	NDCG@5	0.6366	**0.6434 (+1.07%)**
	NDCG@10	0.6182	**0.6219 (+0.6%)**
ML-Latest-Small	F1@5	0.1001	**0.1080 (+7.89%)**
	F1@10	0.1478	**0.1491 (+0.88%)**
	MRR@5	0.3270	**0.3490 (+6.73%)**
	MRR@10	0.3421	**0.3429 (+0.23%)**
	NDCG@5	0.3977	**0.4195 (+5.48%)**
	NDCG@10	0.4439	**0.4480 (+0.92%)**
ML-1M	F1@5	0.1789	**0.1804 (+0.84%)**
	F1@10	0.2457	**0.2473 (+0.65%)**
	MRR@5	0.5984	**0.6014 (+0.50%)**
	MRR@10	0.5603	**0.5625 (+0.39%)**
	NDCG@5	0.6571	**0.6607 (+0.55%)**
	NDCG@10	0.6407	**0.6427 (+0.31%)**
CiaoDVD	F1@5	0.0080	**0.0092 (+15.0%)**
	F1@10	0.0085	**0.0089 (+4.71%)**
	MRR@5	0.0133	**0.0155 (+16.54%)**
	MRR@10	0.0136	**0.0141 (+3.68%)**
	NDCG@5	0.0166	**0.0190 (+14.46%)**
	NDCG@10	0.0204	**0.0212 (+3.92%)**

Table 5 presents the performance comparison between HiGSP and the proposed model when user ratings greater than 2 are used to construct the rating-based adjacency matrix. As shown in the table, the proposed model generally achieved superior performance across all datasets, demonstrating the effectiveness of incorporating user rating information into the recommendation process.

Table 6 presents the performance comparison between HiGSP and the proposed model when user ratings greater than 5 are used to construct the rating-based adjacency matrix [22]. The proposed model also demonstrated overall performance improvements across all datasets.

For the ML-Latest-Small dataset, the final prediction was obtained by combining the outputs of HiGSP and the proposed model with weights of 0.7 and 0.3, respectively. For the ML-1M dataset, the outputs were combined with weights of 0.8 for HiGSP and 0.2 for the proposed model, and the reported performance was measured using these weighted combinations.

V. Conclusions

In this study, we proposed a method to improve the recommendation performance of Graph Signal Processing-based recommender systems by addressing the limitations of the existing HiGSP model [22]. In the original HiGSP model, user–item interactions associated with both high and low user ratings are treated equally when constructing the adjacency matrix, which limits the ability to generate personalized recommendations. To overcome this limitation, we introduced an additional adjacency matrix that incorporates items receiving high user ratings. This approach strengthens the signals of users' preferred items while providing an ensemble effect, thereby enabling more accurate prediction of user preferen

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ces.

To evaluate the effectiveness of the proposed method, experiments were conducted on five benchmark datasets by applying two variants of the proposed model and comparing their performance with that of the original HiGSP model. Experimental results showed that all proposed models, namely Ratings2, Ratings3, Ratings4, and Ratings5, consistently achieved better overall recommendation performance than the original HiGSP model across the five datasets. In particular, the proposed method demonstrated remarkable performance improvements on the Ciao DVD dataset.

An earlier version of this work was presented at a conference [25]. As future work, we plan to address the additional computational cost introduced by using two adjacency matrices in the proposed method. Specifically, we intend to develop a framework that employs a single weighted adjacency matrix, aiming to further improve recommendation accuracy while reducing computational complexity.

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